POWER BI

Product Performance & Ingredient Analysis for Puig's Skin-Care Portfolio

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PROJECT DETAILS

- ***Role*** → Business Intelligence & Data Visualization Analyst (team of two)
- ***Setting*** → University course final project, based on a real case from the Product Testing team at Puig
- ***Technical Stack*** →
  - Power BI Desktop: Power Query, Model view, DAX
  - Custom visuals from AppSource: Radar Chart, Smart Filter Pro
  - Source data: four tables (products, ingredients, product–ingredient links, country scores)
- ***Core Domains*** →
  - Data cleaning & preparation in Power Query
  - Relational data modelling (one-to-many and many-to-many relationships)
  - DAX measures and calculated columns (RANKX, INTERSECT, set comparison)
  - Dynamic reporting: parameters, slicers and Top-N filters
  - Sensory profile visualization with radar charts
  - Dashboard design and user experience

BUSINESS CONTEXT & PROBLEM STATEMENT

Puig is a Spanish multinational in fashion and fragrance. Its Product Testing team studies skin-care products such as creams, oils and serums, both their own and those of competitors. For each product, the team records sensory attributes (for example brightness, stickiness, freshness and skin smoothness), the ingredients it contains, and the score it gets in each country where it is sold.

All of this data existed, but it was split across four separate tables, which made even simple business questions hard to answer. The team needed one visual tool to explore it. Three challenges shaped the project:

- **Real-world, imperfect data:** some values were missing, some numbers were stored with the wrong decimal format, and the sensory attributes used different scales (0–10, 0–100 and open ended). The data was also partly anonymized for confidentiality.
- **Relationships:** one product has many ingredients and is sold in many countries, and one ingredient appears in many products.
- **Non-technical audience:** the users know their products well, but not Power BI. The dashboard had to explain itself, with clear and simple filters.

The objective

🎯 ***Main goal*** → *build an interactive Power BI dashboard that compares Puig products with the competition, to reveal areas for improvement and support business decisions.*

Steps required:

First objective → *clean the four source tables and connect them in one reliable data model.*

Second objective → *answer eight business questions, grouped in three difficulty levels, using dynamic filters, parameters and DAX measures.*

Third objective → *design clear and easy to use pages to prove the numbers.*

DATA & PREPARATION

The final model holds **490 products** (329 from Puig and 161 from competitors) from **83 brands**, **2,129 catalogued ingredients** and **10,191 product-country scores** across **26 countries**.

The four source tables

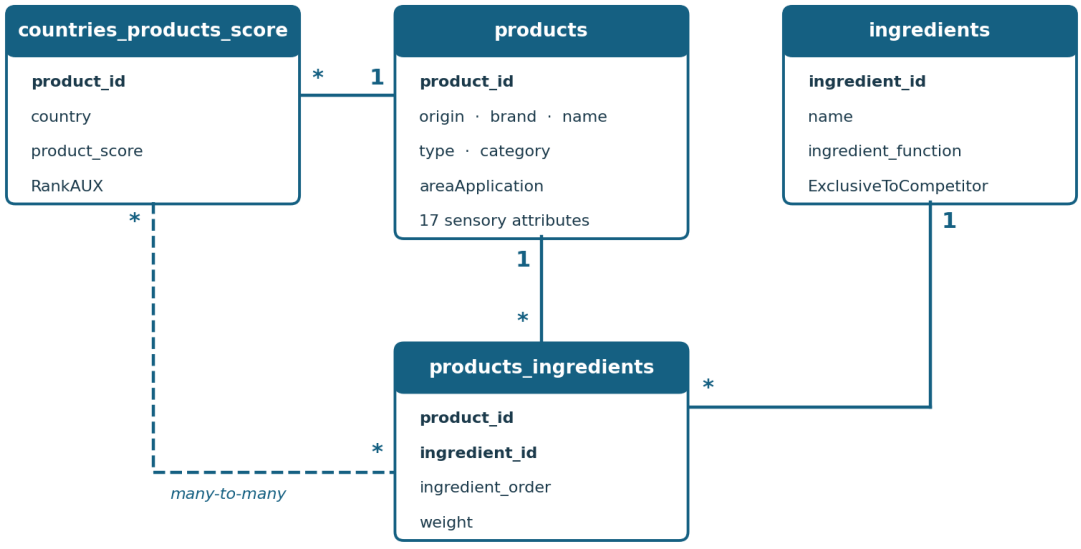
- **Products:** origin (Puig or competitor), brand, name, type, application area and 17 sensory attributes.
- **Ingredients:** ingredient name and function (for example excipient, active or preservative/antioxidant).
- **Products–ingredients:** which ingredients each product contains, in what order, and their weight (Puig products only).
- **Countries–product score:** the 0–10 score of each product in each country.

Cleaning steps in Power Query

1. **Removed low-value columns:** reference, price (almost empty: 408 blank cells vs. 85 filled), dates, quantity, evaluators, packaging and tested-by in Products, plus validated and origin in Ingredients.
2. **Fixed number formats:** scores and sensory attributes came as text with decimal points. They were converted to text, the point was replaced by a comma, and then they were converted to decimal numbers, so every value falls in its correct range.
3. **Put attributes on a common scale:** scaled versions were created for the attributes measured on 0–100 or open-ended scales (silkyRes, waxyRes, greasyRes, oilyRes, easyAbsorption), so all attributes can be compared in one radar chart.
4. **Kept missing categories visible:** empty categories were labelled as not specified, so they appear in the charts instead of silently disappearing.

Data model

The tables are linked through two keys, product_id and ingredient_id. Products connect to both the ingredient links and the country scores, and ingredients connect to the ingredient links. A many-to-many relationship between country scores and ingredient links lets filters flow between countries, products and ingredients.



DASHBOARD STRUCTURE

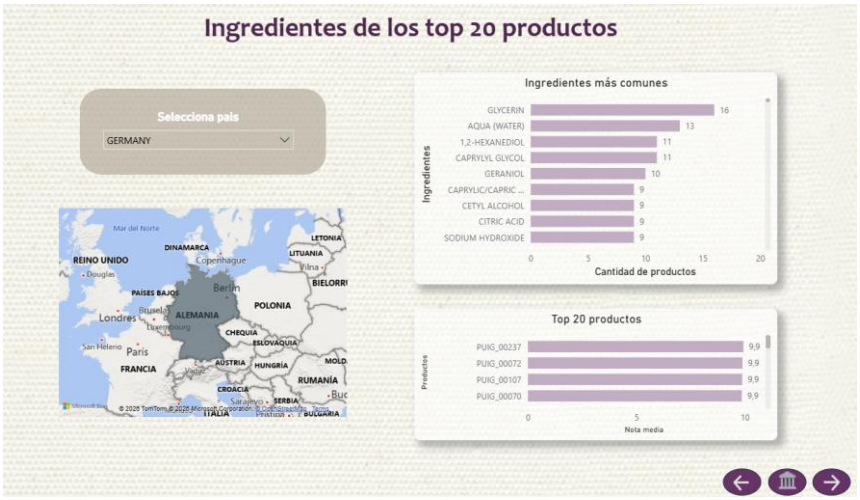
The report has a home page, an index, one page per question and a closing page. Navigation buttons let the user move around without getting lost. Every page follows the same layout, color theme and Puig logo. Extra “check” pages place raw tables next to the charts, so the results can always be verified against the data.

Level	Question answered	Main visuals
L1 · Q1	Which products match my filters, and what do they contain?	Product cards, slicers (origin, brand, area, category, ingredient), ingredient table, KPI counts
L1 · Q2	How is each categorical field distributed?	Bar chart driven by a parameter (user picks the field)
L1 · Q3	What is the sensory profile of one or several products?	Radar chart with a multi-select product slicer
L1 · Q4	Which ingredients are most common in the top 20 products of each country?	Country slicer, map, Top 20 bar chart, ingredient bar chart
L1 · Q5	Which preservative is used most for each product type?	Product type slicer, donut chart, KPI card
L2 · Q1	What is the average sensory profile of each country’s favorite products?	Country slicer, Top-N or minimum-score slicer, radar chart, ranking chart, map, average score
L2 · Q2	Which ingredients do competitors use that Puig does not?	Function slicer, bar chart, KPI card, gauge
L3 · Q1	Which products contain a combination of two ingredients?	Smart Filter Pro slicer, product table, product count, radar chart

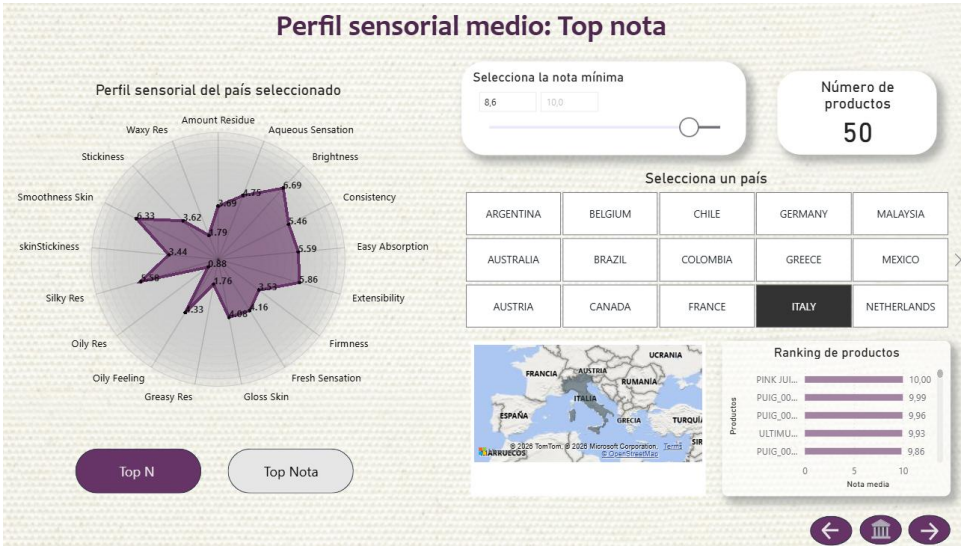
HOW EACH QUESTION WAS SOLVED

Each question was answered with a dedicated page. These are the main technical choices behind them:

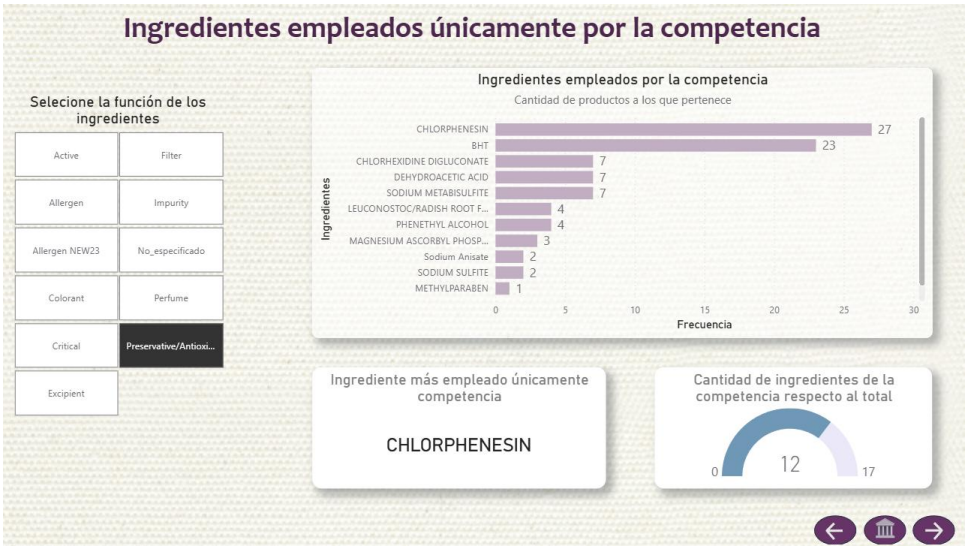
- **Dynamic distribution (L1 · Q2):** a field parameter lets the user choose which field to chart (product, brand, ingredient, function and so on). One chart replaces many, and it updates by itself.
- **Sensory profile (L1 · Q3):** radar charts from AppSource show all 17 attributes as one shape per product. Several radar visuals were tested before choosing the clearest one.
- **Top 20 per country (L1 · Q4):** a calculated column ranks products inside each country with RANKX. The country slicer then shows the best products and the ingredients they share.



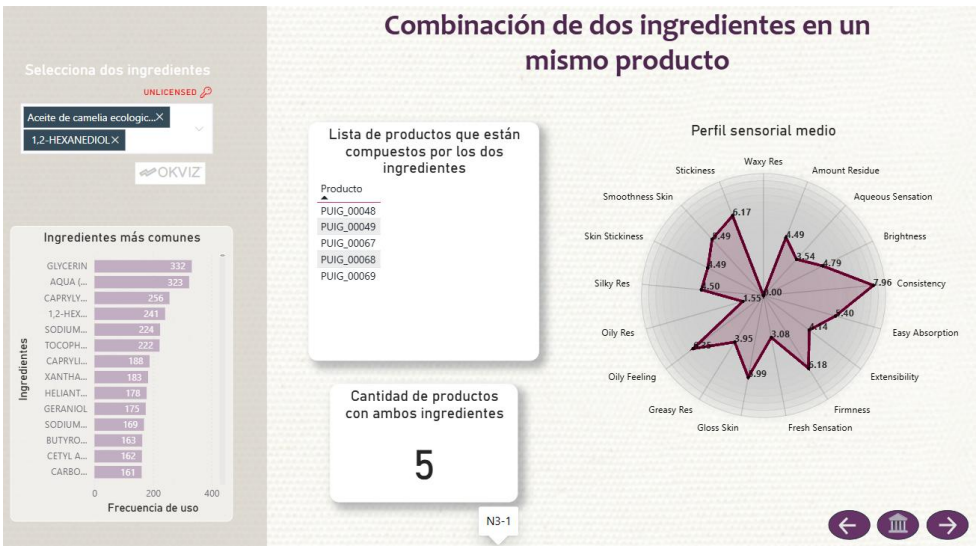
- **Preservatives (L1 · Q5):** only ingredients with the function “Preservative/Antioxidant” are kept. A product type slicer, a donut chart and a KPI card show which one dominates.
- **Flexible Top-N (L2 · Q1):** users can pick the top N products or a minimum score (for example, above 9). Average-profile measures with ALLSELECTED feed one radar chart. Two page versions were built, one by rank and one by score (seen in the image below).



- **Competitor-only ingredients (L2 · Q2):** a calculated column flags every ingredient that no Puig product contains. A function slicer filters them, and a gauge shows their share of the total.



- **Ingredient combinations (L3 · Q1):** a DAX measure uses INTERSECT to keep only products that contain all selected ingredients. Users pick them in the Smart Filter Pro slicer, and the table, counter and radar update at once.



KEY FINDINGS

Using the dashboard, these insights stand out. They should be read carefully, because part of the data was anonymized and 1,404 of the 2,129 ingredients have no recorded function.

- **Portfolio mix:** about two thirds of the products (329 of 490) belong to Puig.
- **Common ingredients:** glycerin (332 products), water (323) and caprylyl glycol (256) are the most frequent ingredients overall.
- **Preservatives:** only 24 preservative ingredients are catalogued. Sodium benzoate (103 products), potassium sorbate (92) and phenoxyethanol (67) dominate, and facial creams use sodium benzoate most often.
- **Ingredient gap:** competitors' products contain 1,267 distinct ingredients versus 400 for Puig, and 1,063 appear only in competitor products. This points to a wider range of formulations on the competitor side, a possible area to explore for new products.

CONCLUSIONS

This project turned four raw tables into an interactive dashboard that answers eight business questions about Puig's skin-care products. The result is easy to use for non-technical teams, flexible thanks to parameters and dynamic filters, and reliable thanks to validation pages. The main lessons were that careful data preparation and modelling matter as much as the visuals, and that DAX measures make a dashboard truly dynamic.